



# Modulating the surface amphipathicity of Ni-Sn composite nanoparticles for efficient upgrading of aqueous ethanol to C<sub>6+</sub> alcohols

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## ABSTRACT

Aqueous ethanol upgrading has attracted great attention for producing high-valued higher alcohols. Nevertheless, the chain growth for higher alcohol production involving water–oil biphasic reaction, is usually limited by the traditional single-phase (hydrophilic or lipophilic) catalyst, leading to poor selectivity of C<sub>6+</sub> alcohols. Herein, we demonstrate that modulating the surface amphipathicity of the active Ni-Sn composite nanoparticles by using hybrid carbon shell and TiO<sub>2</sub> support significantly boosted its catalytic performance. A high ethanol conversion of 70.6 % and excellent C<sub>6+</sub> alcohols selectivity of 57.5 % was achieved. Deep studies show that the enhanced catalytic performance was mainly originated from the facilitated emulsion catalysis by the amphiphilic NiSn@C-TiO<sub>2</sub> catalyst, by which the reaction process from aqueous ethanol self-coupling to intermolecular cross-coupling at the oil/water interface was dramatically accelerate. This work developed a novel emulsion catalyst for water/oil biphasic interface catalysis, approaching highly efficient production of high-valued higher alcohols from direct upgrading of aqueous ethanol.

## 1. Introduction

In view of the world's universal call for sustainable development and low-carbon society, biomass has been regarded as a promising candidate for producing bio-chemicals to replace the conventional petroleum refining products (Goh et al., 2022; He et al., 2023; Karpagam et al., 2020; Luo et al., 2023a; Mika et al., 2018; Morganti et al., 2018; Zhao et al., 2020). Over the past decades, substantial efforts have been devoted to biomass upgrading. The catalytic conversion of biomass to ethanol has attracted great interests of both academics and industry, since the bio-ethanol derived from biomass fermentation can serve as a platform chemical for the generation of other value-added chemicals (Han et al., 2019; Luo et al., 2023b; Pang et al., 2021; Pérez et al., 2011; Shields and Boopathy, 2011; Wu et al., 2021a; Zhang et al., 2023). Especially, higher alcohols with carbon numbers  $\geq 6$  (C<sub>6+</sub> alcohols) have been recognized as promising fine chemicals to produce bio-medical materials and bio-jet fuel due to their unique molecular structure and higher energy density (Wang et al., 2020a; Wang et al., 2020b; Yuan et al., 2022; Zhang et al., 2016). As reported by Liao et al. jet fuel hydrocarbons of 56.7 % yield obtained from C<sub>8</sub>-C<sub>16</sub> alcohols

hydrodeoxygenation over Co-Mo<sub>2</sub>C catalyst is mainly C<sub>8</sub>-C<sub>16</sub> branched alkanes (Liao et al., 2022). Moreover, C<sub>6+</sub> alcohols can be directly blended with diesel fuel due to its phase compatibility with diesel (De Poures et al., 2020; Rorrer et al., 2018). Nevertheless, the quest for a highly efficient catalyst for ethanol upgrading to C<sub>6+</sub> alcohols is driven by the current drawbacks including low ethanol conversion and poor selectivity to C<sub>6+</sub> alcohols (Earley et al., 2015; Galadima and Muraza, 2015; Pang et al., 2016).

In principle, ethanol can be converted into higher alcohols through the sequential aldol condensation, dehydration, and hydrogenation via the classic Guerbet route (Eagan et al., 2019; Han et al., 2021; Wu et al., 2021b). Over the past decades, great attentions have been paid on the upgrading of ethanol to butanol. For instance, organic Mn-based complexes were reported as homogeneous catalysts for ethanol upgrading with *n*-butanol selectivity up to 92 % (Fu et al., 2017). Kulkarni and co-workers developed organic Mn-based pincer complexes for upgrading of ethanol yielding 34 % of *n*-butanol (Kulkarni et al., 2018). However, homogeneous catalysts for ethanol upgrading face challenges of difficult separation and recycle, high cost and poor selectivity of C<sub>6+</sub> alcohols (Aitchison et al., 2016; Liu et al., 2019; Zhang et al., 2016). As a result,

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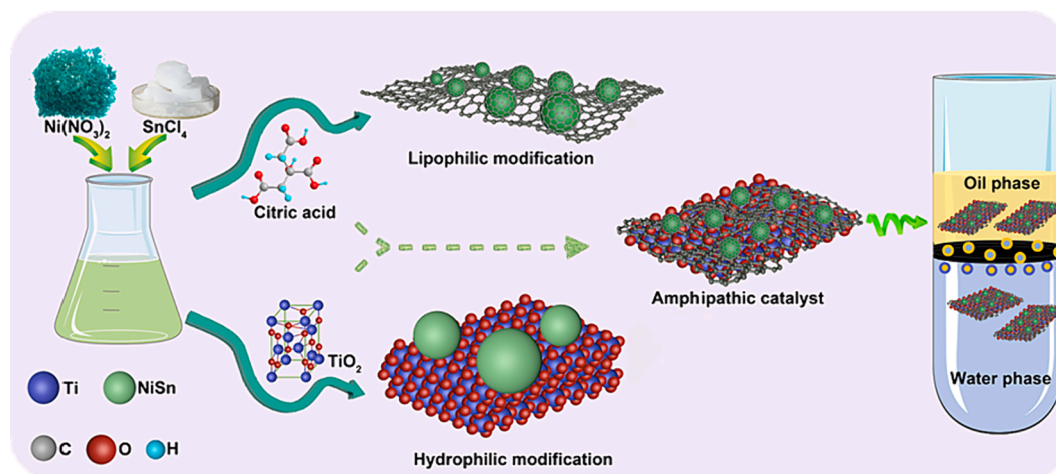
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**Scheme 1.** Schematic illustration for the modulation of surface hydrophilic and lipophilic properties of Ni-Sn composite catalyst.

heterogeneous catalysis instead has been recently studied for ethanol upgrading (Kozłowski and Davis, 2013). Nevertheless, the heterogeneous catalysts developed are far away from satisfaction due to their low activity and poor selectivity for  $C_6+$  alcohols.

As ethanol is hydrophilic while  $C_6+$  alcohols are lipophilic, aqueous ethanol upgrading for  $C_6+$  higher alcohols necessarily involves biphasic catalysis for transforming ethanol in water phase to  $C_6+$  alcohols oil phase. To improve the catalytic performance of ethanol upgrading to  $C_6+$  alcohols, one needs to modulate the distribution of catalyst in the water–oil biphasic system (Crossley et al., 2010). In this work, we precisely modulated the lipophilic and hydrophilic properties of a active Ni-Sn composite catalyst through surface modification with hybrid support of carbon shell and  $TiO_2$ . Consequently, the amphiphilic catalyst exhibited the properties of emulsion catalysis, which can firstly catalyze the self-coupling of aqueous ethanol and then crossing-coupling of intermolecular at oil/water interface, leading to the significant enhancements on ethanol conversion and  $C_6+$  alcohols selectivity. Impressively, the optimal catalyst displays the  $C_6+$  alcohols selectivity of 57.5 % with ethanol conversion of 70.6 %, being the best results for ethanol upgrading to  $C_6+$  alcohols. This work provides an efficient strategy for the  $C_6+$  alcohols production by direct upgrading of aqueous ethanol from biomass fermentation.

## 2. Experimental section

### 2.1. Catalyst preparation

#### 2.1.1. Preparation of NiSn@C/ $TiO_2$

NiSn@C/ $TiO_2$  catalysts were prepared by impregnation method followed by calcination. In a typical synthetic procedure for NiSn@C/ $TiO_2$ -1 with a C/ $TiO_2$  molar ratio of 1, 6.36 g  $Ni(NO_3)_2 \cdot 6H_2O$ , 2.93 g  $TiO_2$  (~150 nm, anatase, hydrophilic), 0.44 g  $SnCl_4 \cdot 5H_2O$  and 7.71 g citric acid monohydrate were dissolved in deionized water (60 mL) under continuous stirring to obtain a homogeneous solution. After vigorously stirring at room temperature for 2 h, the mixture was dried at 100 °C for 48 h in an oven. Finally, the resultant precursor was calcined in Ar flow (20 mL/min) at 550 °C for 3 h. Other NiSn@C/ $TiO_2$  were prepared by altering the molar ratio of citric acid and  $TiO_2$ .

#### 2.1.2. Preparation of NiSn/ $TiO_2$

NiSn/ $TiO_2$  was synthesized through a classic impregnation method. Firstly, 6.36 g  $Ni(NO_3)_2 \cdot 6H_2O$ , 2.93 g  $TiO_2$  and 0.44 g  $SnCl_4 \cdot 5H_2O$  were dissolved in deionized water (60 mL) in sequence under continuous stirring to obtain a homogeneous solution. After vigorously stirring at room temperature for 12 h, the mixture was dried at 100 °C for 24 h in an oven. Finally, the resultant precursor was reduced by  $H_2$  (40 mL/min)

at 550 °C for 3 h, which was denoted as NiSn/ $TiO_2$ .

#### 2.1.3. Preparation of NiSn@C

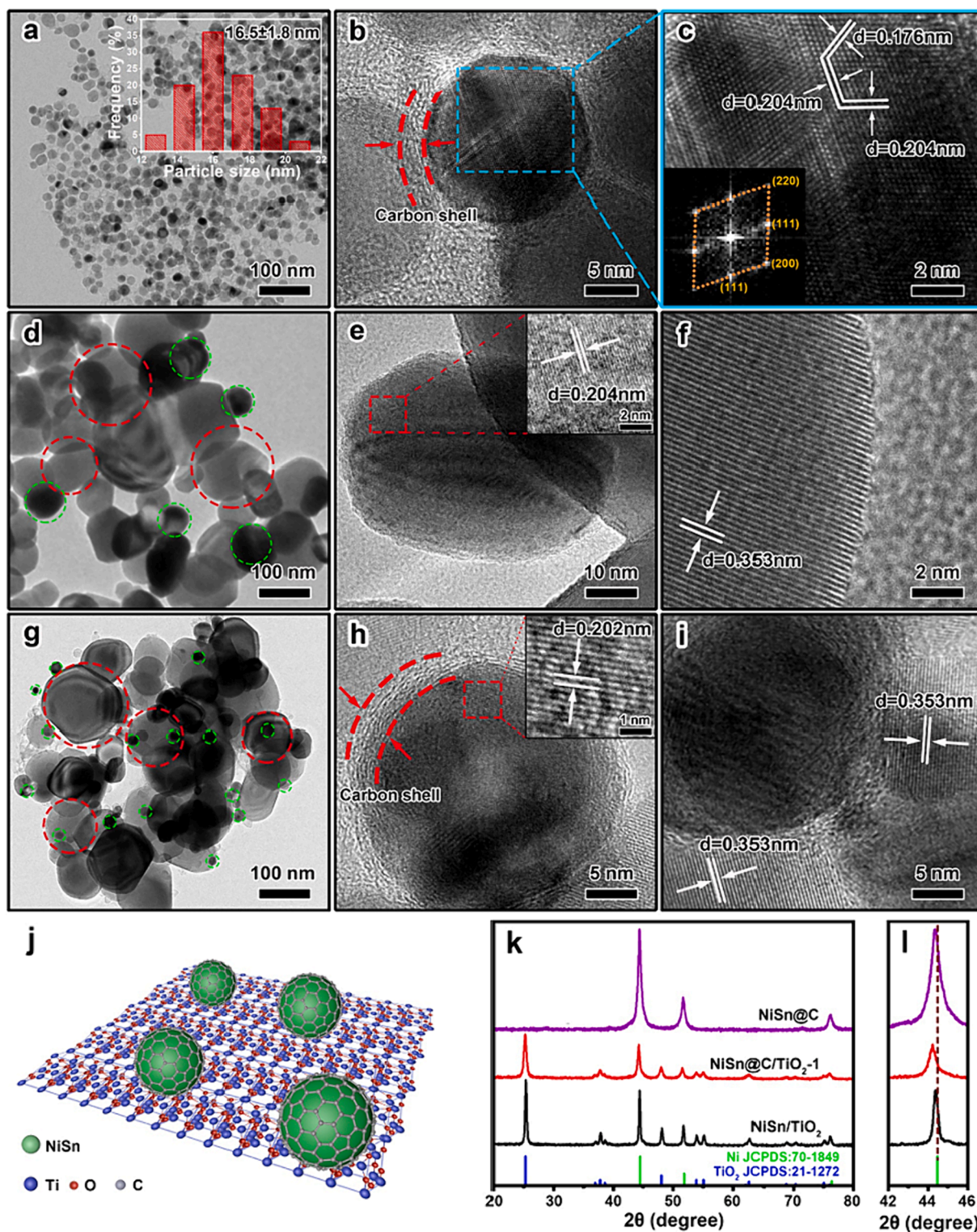
NiSn@C was synthesized by a facile sol-gel and one-step pyrolysis/carbonization strategy. Firstly, 6.36 g  $Ni(NO_3)_2 \cdot 6H_2O$ , 7.71 g citric acid monohydrate and 0.44 g  $SnCl_4 \cdot 5H_2O$  were dissolved in deionized water (60 mL) under continuous stirring to obtain a homogeneous solution. After vigorously stirring at room temperature for 2 h, the mixture was gradually gelled by aging at 100 °C and subsequently dried at 100 °C for 48 h in an oven. Finally, the resultant precursor was calcined in Ar flow (20 mL/min) at 550 °C for 3 h.

### 2.2. Catalyst characterization

X-ray diffraction (XRD) measurement was performed on a MiniFlex 600 diffractometer using Cu K radiation with a step size of 0.0167° at 10° per minute in the 2 range of 5–80°. Scanning electron microscope (SEM) was conducted on a S-4800 microscope (HITACHI Corp., Tokyo, Japan). Transmission electron microscopy (TEM) and high resolution transmission electron microscopy (HRTEM) images were collected using a JEM-2100F microscope (JEOL Corp., Tokyo, Japan) operated at 120 kV. Before analysis, the sample was dispersed in ethanol by ultrasonication and then mounted on C-coated Cu grids, followed by evaporation of the solvent at 60 °C. X-ray photoelectron spectroscopy (XPS) for analyzing the surface properties of catalysts was conducted on a Thermo Fisher Scientific Escalab 250 Xi photoelectron spectrometer and values were revised by C 1 s (284.8 eV). The water-droplet contact angle was measured on a Dataphysics OCA100 instrument (Detailed method is shown in the Supporting Information). Before measuring the emulsion properties, the products contained NiSn@C/ $TiO_2$ -1 catalyst obtained under optimal conditions was sonicated and vibrated for 1 h. The droplet of emulsion was characterized by a Polarizing Microscope. The extended X-ray absorption measurements were carried out on the sample at 21A X-ray nanodiffraction beamline of Taiwan Photon Source (TPS), National Synchrotron Radiation Research Center (NSRRC). This beamline adopted 4-bounce channel-cut Si (1 1 1) monochromator for mono-beam X-ray nanodiffraction and X-ray absorption spectroscopy. The end-station equipped with three ionization chambers and Lytle/SDD detector after the focusing position of KB mirror for transmission and fluorescence mode X-ray absorption spectroscopy. The photon flux on the sample is range from  $1 \times 10^{11} \sim 3 \times 10^9$  photon/sec for X-ray energy from 6 to 27 keV.

### 2.3. Aqueous ethanol upgrading

Ethanol upgrading was performed in a 50 mL stainless-steel batch



**Fig. 1.** (a) TEM image and size distribution of Ni-Sn nanoparticles. (b) TEM image with high magnification and (c) HRTEM image of NiSn@C catalyst. Inset in (c) is the corresponding pattern after fast Fourier transform. (d, e) TEM images and (f) HRTEM image of NiSn/TiO<sub>2</sub> catalyst. Inset of (e) is the HRTEM image at the selected area. (g, h) TEM images and (i) HRTEM image of NiSn@C/TiO<sub>2</sub> catalyst. Inset of (h) is the HRTEM image collected at the selected area. (j) Scheme of NiSn@C/TiO<sub>2</sub>. (k, l) XRD patterns of various catalysts.

reactor (Fig. S1: MS-50-316L, Anhui CHEM<sup>N</sup> Instrument Co., Ltd, Hefei, China). Typically, 0.5 g NaOH and 0.5 g catalyst were added into the mixture of 10.0 g ethanol and 10.0 g water. Afterwards, the reactor was flushed with H<sub>2</sub> for 5 times and maintained in H<sub>2</sub> at a total pressure of 0.1 MPa. Finally, the reactor was heated to the target temperature

with a heating rate of 4 °C/min before a 18 h hold, with the continuously stirring rate of 500 rpm. After reaction, the reactor was cooled down to room temperature. The product detection and collection mainly consist of gas product and liquid product. Gas sample was collected with the air bags, and the liquid sample obtained by centrifugation was weighed and

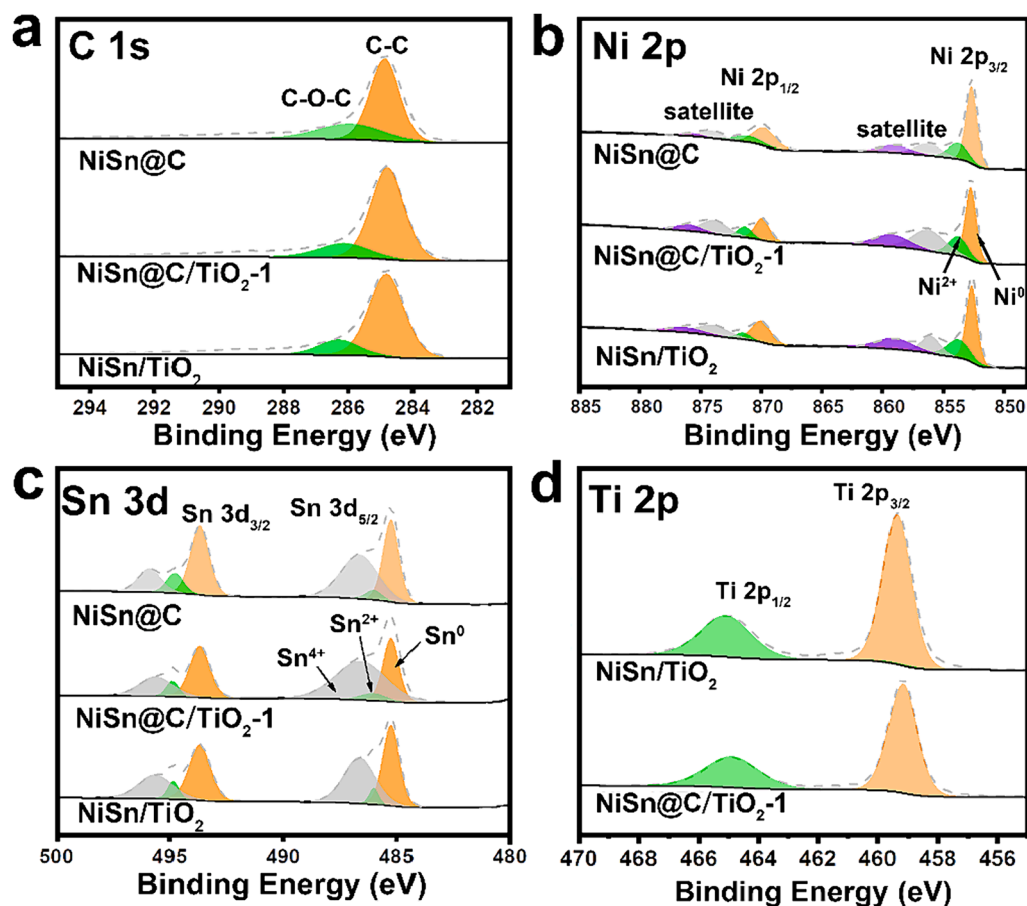


Fig. 2. XPS spectra of NiSn/TiO<sub>2</sub>, NiSn@C and NiSn@C/TiO<sub>2</sub>-1: (a) C 1 s spectra, (b) Ni 2p spectra, (c) Sn 3d spectra, and (d) Ti 2p spectra.

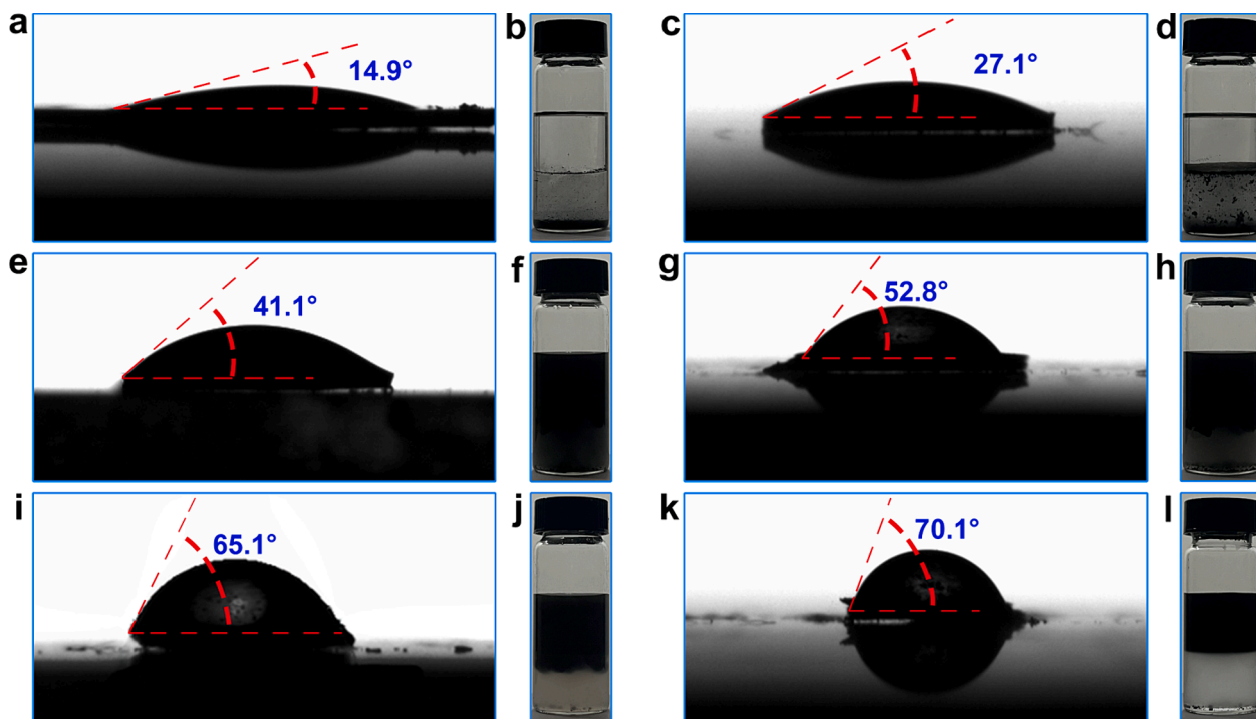
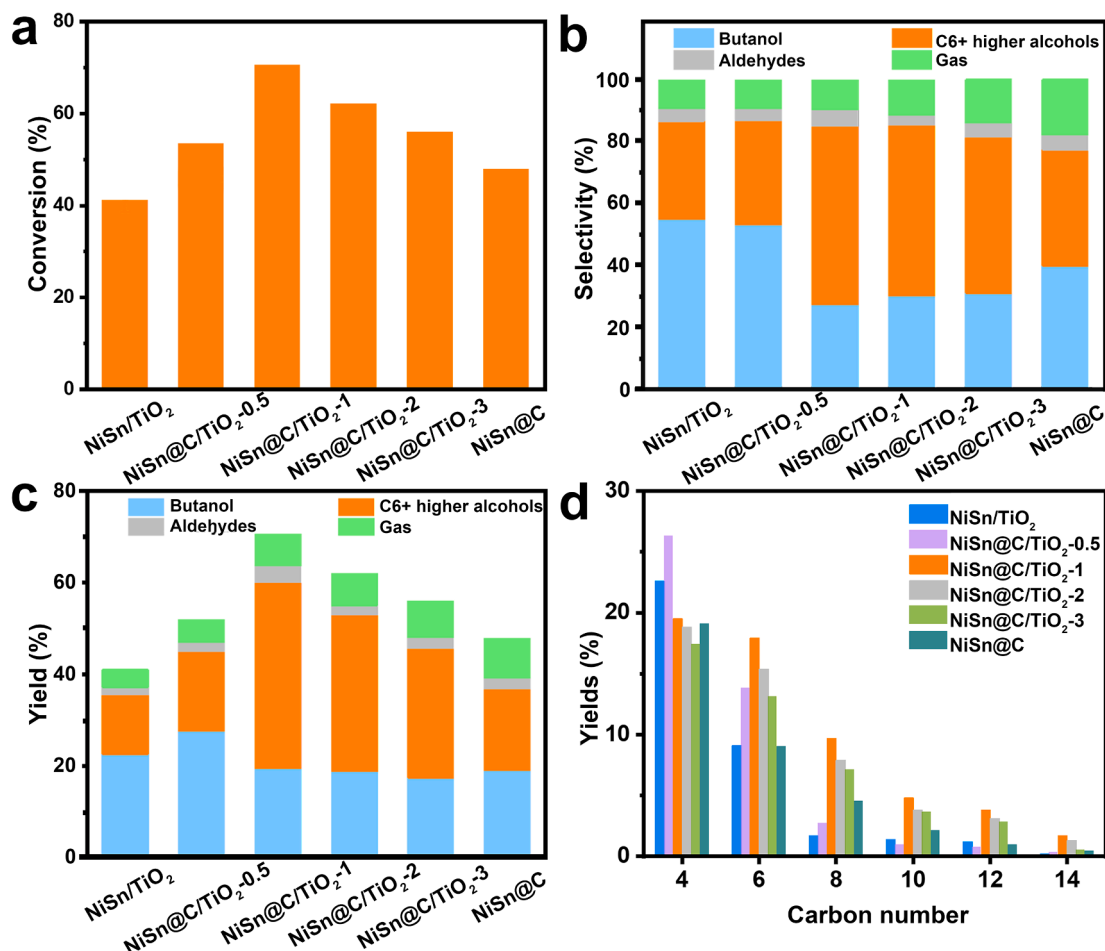


Fig. 3. The water-droplet contact angle tests for (a) NiSn/TiO<sub>2</sub>, (c) NiSn@C/TiO<sub>2</sub>-0.5, (e) NiSn@C/TiO<sub>2</sub>-1, (g) NiSn@C/TiO<sub>2</sub>-2, (i) NiSn@C/TiO<sub>2</sub>-3, and (k) NiSn@C. Distributions of various catalysts in water/octanol mixture: (b) NiSn/TiO<sub>2</sub>, (d) NiSn@C/TiO<sub>2</sub>-0.5, (f) NiSn@C/TiO<sub>2</sub>-1, (h) NiSn@C/TiO<sub>2</sub>-2, (j) NiSn@C/TiO<sub>2</sub>-3, and (l) NiSn@C.



**Fig. 4.** Catalytic performance of ethanol upgrading. (a) Ethanol conversion, (b) selectivity of products, (c) yields of products, and (d) distributions of higher alcohols for various catalysts. Reaction conditions:  $m_{\text{NaOH}} = 0.5$  g,  $m_{\text{catalyst}} = 0.5$  g,  $m_{\text{ethanol}} = 10$  g,  $m_{\text{water}} = 10$  g,  $T = 250$  °C, and  $t = 18$  h.

used for subsequent analysis. Note that the Ni-based catalyst provides metallic sites and NaOH provides basic sites.

The detailed product analysis method can be found in the [Supporting Information](#). The conversion, yield, selectivity and distribution of the products were calculated by the following equations:

$$\text{Conversion (C)} = \frac{\text{Moles of converted ethanol}}{\text{Moles of the inlet ethanol}} \times 100\% \quad (1)$$

$$\text{Selectivity (S)} = \frac{\text{Moles of carbon in the target product}}{\text{Moles of carbon in all products}} \times 100\% \quad (2)$$

$$\text{Yield (Y)} = \frac{\text{Moles of carbon in the product}}{\text{Moles of carbon in ethanol fed to the reactor}} \times 100\% \quad (3)$$

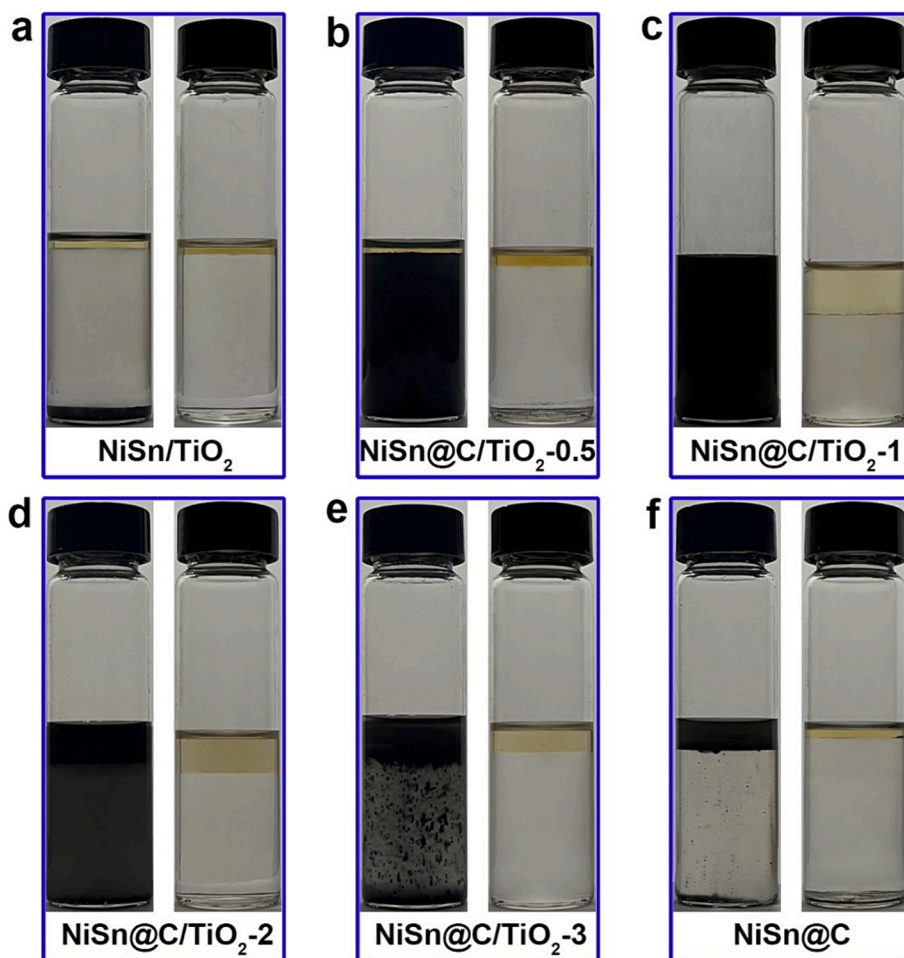
### 3. Results and discussion

#### 3.1. Structure of the catalysts

NiSn@C/TiO<sub>2</sub> was prepared via a chemical approach by in situ assembling active Ni-Sn nanoparticles into a hybrid support constituting lipophilic carbon shell and hydrophilic TiO<sub>2</sub> (see detailed information in experimental section). In particular, citric acid was used as reductant, surface ligand, as well as the precursor for the carbonation of Ni-Sn nanoparticles, while TiO<sub>2</sub> was used directly to modify the hydrophilia of the catalyst. NiSn@C/TiO<sub>2</sub> was supposed to be amphiphilic due to the hybrid support (Scheme 1). Here, NiSn@C, NiSn/TiO<sub>2</sub>, and NiSn@C/TiO<sub>2</sub> with a C/TiO<sub>2</sub> molar ratio of 1 (namely NiSn@C/TiO<sub>2</sub>-1) were selected as examples (Fig. 1). For NiSn@C, transmission electron

microscopy (TEM) image displays that the mean size of Ni-Sn nanoparticles is 16.5 nm (Fig. 1a). TEM image with high magnification implies that the Ni-Sn nanoparticles are coated with carbon shell (Fig. 1b). The lattice distances of 0.176 and 0.204 nm in the high-resolution TEM (HRTEM) image are ascribed to Ni(200) and Ni(201) facet, respectively (Fig. 1c). Note that the measured lattice distances of (200) and (201) facets are slightly larger than the standard values, indicating the expansion of lattice caused by Sn introduction. For NiSn/TiO<sub>2</sub>, Ni-Sn nanoparticles with an average size of ~ 60 nm were loaded on TiO<sub>2</sub>, which were labeled with green and red circles, respectively (Fig. 1d). The lattice distances of 0.204 and 0.353 nm can be indexed as the (111) facet of Ni and the (101) facet of TiO<sub>2</sub>, respectively (Fig. 1e, f). For NiSn@C/TiO<sub>2</sub>-1, Ni-Sn nanoparticles coated with carbon shell (green circle in Fig. 1g) were loaded on TiO<sub>2</sub> (red circle in Fig. 1g). Similarly, the lattice distances of 0.202 and 0.353 nm are ascribed to the (111) facet of Ni and the (101) facet of TiO<sub>2</sub>, respectively (Fig. 1h, i). Consequently, a scheme for the structure of NiSn@C/TiO<sub>2</sub>-1 is given in Fig. 1j.

The BET surface area of NiSn@C/TiO<sub>2</sub> catalysts with various controlled C/TiO<sub>2</sub> ratios was demonstrated in Table S1. The NiSn@C catalyst had the highest BET surface area of 115 m<sup>2</sup>/g<sub>cat</sub>. Surprisingly, the BET surface area of NiSn/TiO<sub>2</sub> catalyst reached 49 m<sup>2</sup>/g<sub>cat</sub>, which is much larger than that of 10 m<sup>2</sup>/g<sub>cat</sub> for NiSn@C/TiO<sub>2</sub>-0.5 catalyst. This is because when a small amount of C is incorporated into the TiO<sub>2</sub> support, C will enter the pore of TiO<sub>2</sub> support and block part of the pores. X-ray diffraction (XRD) measurement was performed to study the structures of the catalysts. The peaks at 44.4°, 51.8° and 76.1° in the XRD patterns of NiSn@C, NiSn/TiO<sub>2</sub>, and NiSn@C/TiO<sub>2</sub>-1 are ascribed to Ni (111), Ni (200) and Ni (220) facets, respectively (JCPDS No. 70-



**Fig. 5.** The pictures of reaction solutions after aqueous ethanol upgrading with and without catalyst: (a) NiSn/TiO<sub>2</sub>, (b) NiSn@C/TiO<sub>2</sub>-0.5, (c) NiSn@C/TiO<sub>2</sub>-1, (d) NiSn@C/TiO<sub>2</sub>-2, (e) NiSn@C/TiO<sub>2</sub>-3, and (f) NiSn@C. (Left: with catalyst, right: without catalyst).

1849) (Fig. 1k). The slight negative shift of the peak corresponding to Ni (111) facet is attributed to the introduction of Sn (Fig. 1l), which is consistent with the result of lattice expansion from HRTEM images. Moreover, the peaks at 25.1°, 37.4°, 47.8°, 53.3°, 54.7°, 62.2°, 68.1°, 69.8°, 75.5° are corresponded to TiO<sub>2</sub> (101), (004), (200), (105), (211), (204), (116), (220), (301) facets, respectively (JCPDS No. 21-1272). In addition to the peaks of Ni and TiO<sub>2</sub>, no obvious changes were observed in the XRD patterns when C/TiO<sub>2</sub> ratio was increased from 0.5 to 3 (Fig. S3), indicating that the carbon shell is amorphous.

Furthermore, the surface properties of the catalysts were measured by X-ray photoelectron spectroscopy (XPS). In C 1 s XPS spectra, the peaks at 284.8 and 285.9 eV are ascribed to C–C and C–O–C, respectively (Fig. 2a). Ni mainly consists of Ni<sup>0</sup> with partial oxidation on the surface, as revealed by the presences of peaks at 852.6 and 853.7 eV in Ni 2p XPS spectra (Fig. 2b and Table S2) (Usha et al., 2022; Zhu et al., 2020). In the Sn 3d<sub>5/2</sub> XPS spectra, the broad peak can be deconvoluted into three peaks, which correspond to Sn<sup>0</sup>, Sn<sup>2+</sup>, and Sn<sup>4+</sup>, respectively (Fig. 2c and Table S3) (Baggetto et al., 2013). Besides, the results from Ti 2p XPS spectra suggest that Ti in NiSn/TiO<sub>2</sub> and NiSn@C/TiO<sub>2</sub>-1 presents as the oxidation state (Ti<sup>4+</sup>) (Fig. 2d).

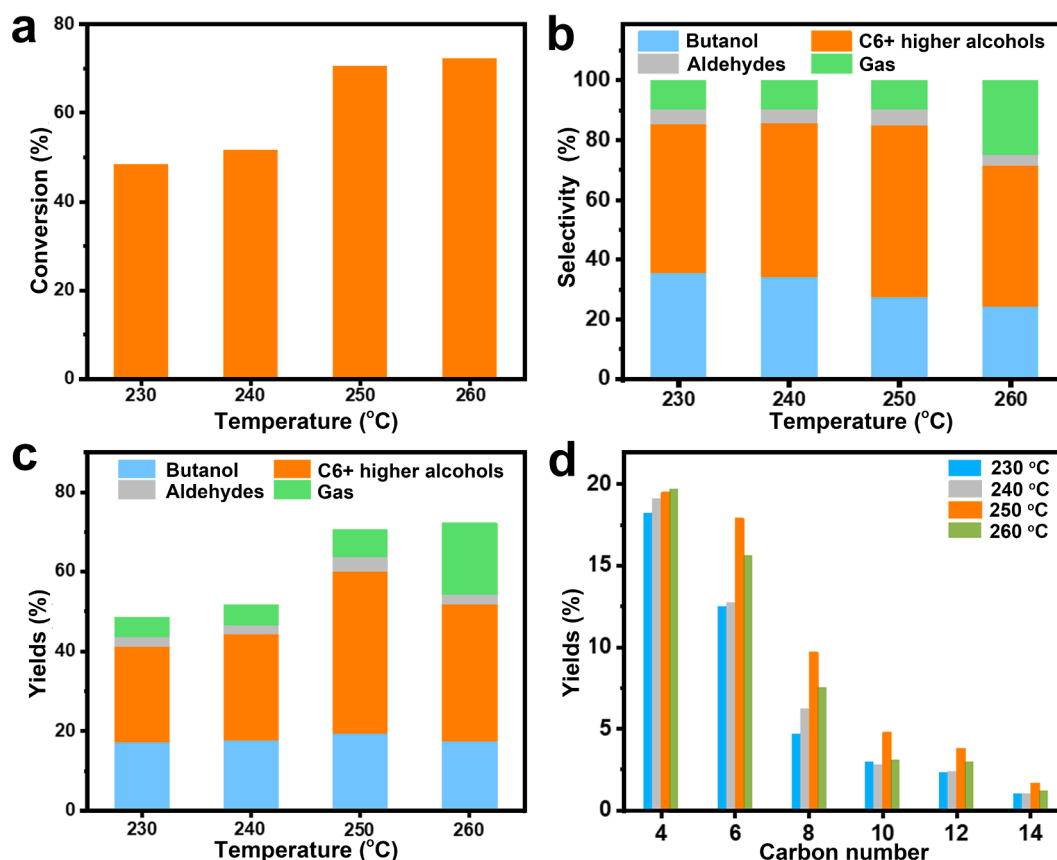
### 3.2. The hydrophilicity and lipophilicity of the catalysts

Considering the co-existence of the lipophilic carbon shell and the hydrophilic TiO<sub>2</sub>, the amphipathic properties of the NiSn@C/TiO<sub>2</sub> should be strongly related to the molar ratio of C/TiO<sub>2</sub>. The water-droplet contact angle measurement was thus conducted to evaluate

the hydrophilic and hydrophobic performance of the NiSn@C/TiO<sub>2</sub> catalysts with different C/TiO<sub>2</sub> ratios. As shown in Fig. 3a, the contact angle of NiSn/TiO<sub>2</sub> is as low as 14.9°, suggesting that the introduction of TiO<sub>2</sub> can significantly improve the hydrophilicity of Ni-Sn composite nanoparticles. After adding NiSn/TiO<sub>2</sub> into the mixture of water (bottom layer) and octanol (upper layer), it is observed that NiSn/TiO<sub>2</sub> is almost distributed in the water phase (Fig. 3b). Moreover, the contact angle significantly increases when the Ni-Sn nanoparticles was decorated with carbon shell. For NiSn@C/TiO<sub>2</sub>-0.5 with a C/TiO<sub>2</sub> ratio of 0.5, the contact angle increases to 27.1° (Fig. 3c), leading to the re-distribution of sample both in the water phase and at the water–oil interface (Fig. 3d). The contact angle further increases to 41.1° when the molar ratio of C/TiO<sub>2</sub> was increased to 1 (NiSn@C/TiO<sub>2</sub>-1), resulting in the uniform distribution of the sample in water–oil interface (Fig. 3e,f). As increasing the C/TiO<sub>2</sub> molar ratio from 1 to 3, the contact angle further increases from 41.1° to 65.1°, leading to the distribution of the sample in oil phase (Fig. 3e–j). Typically, for NiSn@C catalyst without TiO<sub>2</sub>, the contact angle is 70.1° and the sample almost completely distributes in the oil phase (Fig. 3k,l). The above results clearly demonstrate that the lipophilic and hydrophilic properties of catalysts can be precisely modulated by altering the molar ratio of C/TiO<sub>2</sub>.

### 3.3. Aqueous ethanol upgrading

As mentioned above, the aqueous ethanol upgrading involves phase transfer catalysis once the C<sub>6+</sub> alcohols were formed. The amphiphilic NiSn@C/TiO<sub>2</sub> with different C/TiO<sub>2</sub> ratios were tested for aqueous



**Fig. 6.** (a) Ethanol conversion, (b) selectivity of liquid products, (c) yield of liquid products, and (d) distribution of liquid products over NiSn/C/TiO<sub>2</sub>-1 during aqueous ethanol upgradation. Reaction conditions:  $m_{\text{NaOH}} = 0.5$  g,  $m_{\text{catalyst}} = 0.5$  g,  $m_{\text{ethanol}} = 10$  g,  $m_{\text{water}} = 10$  g,  $T = 230\text{--}260$  °C, and  $t = 18$  h.

ethanol upgrading to explore their biphasic catalytic performances. As shown in Fig. 4a, the ethanol conversions are 41.1 % and 47.9 % for NiSn/TiO<sub>2</sub> and NiSn@C, respectively. A volcano shape of ethanol conversion was observed when the C/TiO<sub>2</sub> ratio was increased from 0.5 to 3 (Fig. 4a). Moreover, the selectivity of C<sub>6</sub> + alcohol displays a similar tendency to the yield with the increase of C/TiO<sub>2</sub> ratio (Fig. 4b). The optimal NiSn@C/TiO<sub>2</sub>-1 exhibits the best performance with ethanol conversion of 70.6 % and C<sub>6</sub>+ alcohols selectivity of 57.5 %, which gives a total C<sub>6</sub>+ alcohols yield of 40.6 % (Fig. 4c). The distribution of higher alcohols with different carbon numbers (Fig. 4d) also clearly proved the superior performance of the NiSn@C/TiO<sub>2</sub>-1 catalyst.

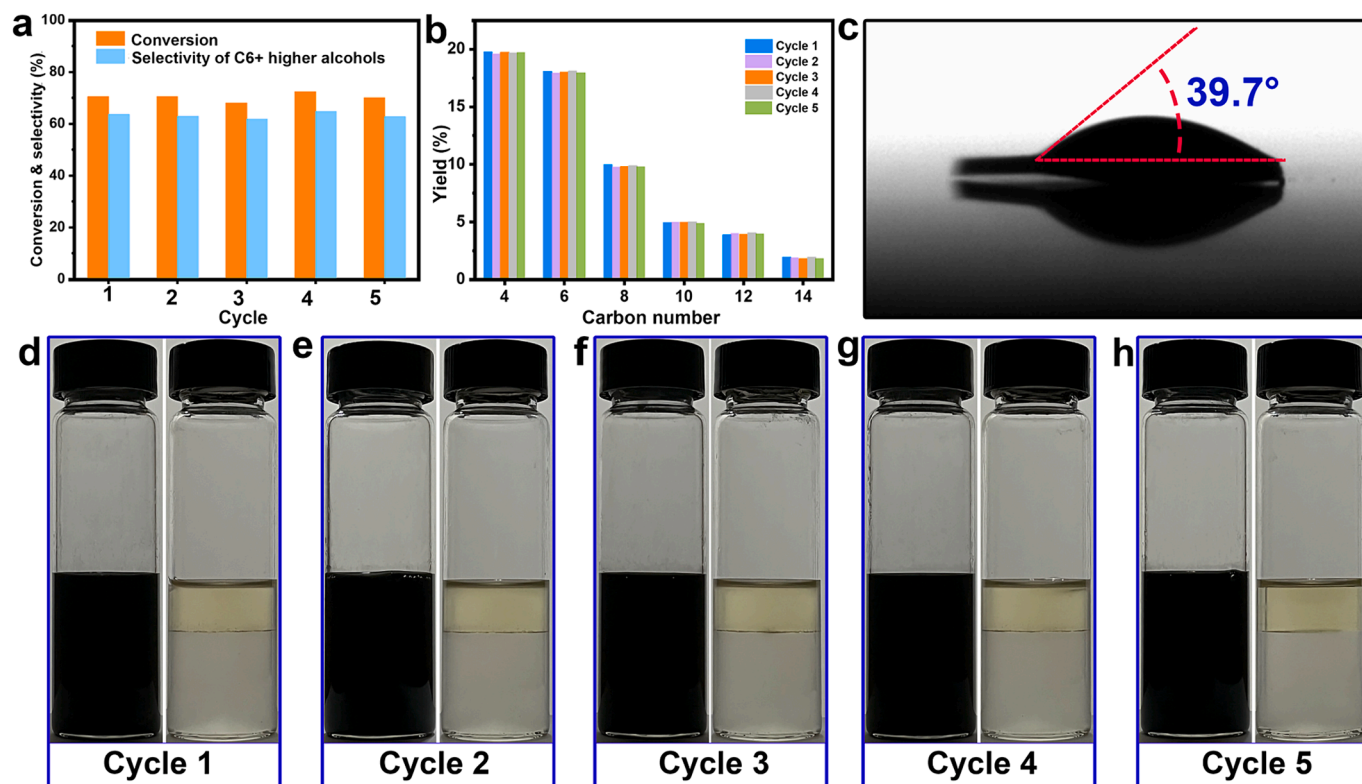
The distribution of the catalysts in the biphasic reaction solution after ethanol upgradation reaction is shown in Fig. 5. It was found that the NiSn/TiO<sub>2</sub> catalyst was deposited at the bottom of water phase and NiSn/C catalyst was in the oil phase. However, NiSn@C/TiO<sub>2</sub> catalysts were distributed both in the water phase and oil phase. Interestingly, NiSn@C/TiO<sub>2</sub>-1 showed uniform distribution in the oil and water phases. After separating the NiSn@C/TiO<sub>2</sub>-1 catalyst via centrifugation, two individual phases (water phase and oil phase) were clearly observed, which indicated that the amphipathicity of NiSn@C/TiO<sub>2</sub>-1 catalyst promoted the formation of oil/water emulsion phase. Note that NiSn@C/TiO<sub>2</sub>-1 catalyst at the oil/water interface exhibited the catalytic effect of emulsion, which enhanced the crossing-coupling of ethanol and C<sub>4</sub>+ alcohols to form C<sub>6</sub>+ alcohols.

Moreover, the effect of reaction temperature on ethanol upgrading over the NiSn@C/TiO<sub>2</sub>-1 was investigated. As shown in Fig. 6a, the ethanol conversion gradually increased from 48.4 % to 72.2 % when reaction temperature was increased from 230 to 260 °C. High temperature favors the ethanol conversion. Nevertheless, the selectivity of gaseous byproducts gradually increases with the temperature increase due to the aqueous phase reforming reaction. For instance, when the

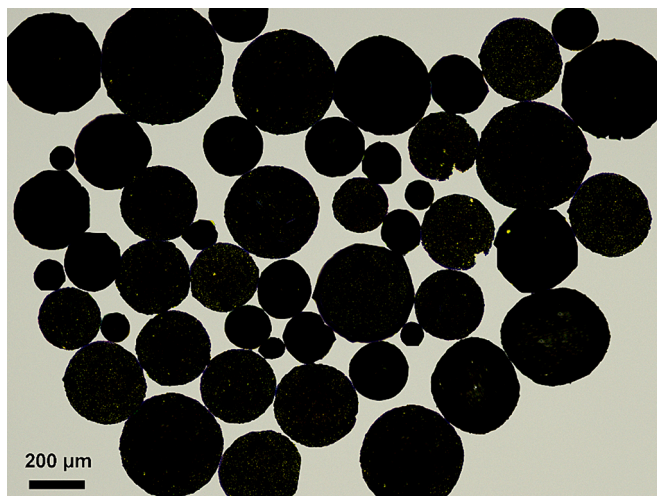
temperature was 230 °C, the selectivities of liquid products and gaseous byproducts were 90.3 % and 9.7 %, respectively. When the reaction temperature was increased to 260 °C, the selectivity of gaseous byproducts obviously increased to 24.8 %, while the selectivity of liquid products decreased to 75.2 % (Fig. 6b). Correspondingly, the highest C<sub>6</sub>+ alcohols yield of 40.6 % was obtained at the temperature of 250 °C (Fig. 6c).

Additionally, the stability of NiSn@C/TiO<sub>2</sub>-1 catalyst was further evaluated by 5 consecutive cycles under the optimized reaction conditions. As shown in Fig. 7a, the ethanol conversion and C<sub>6</sub>+ alcohols selectivity maintained at about 70 % and 62 % for all the five cycle runs. Besides, no obvious changes in the carbon number distribution of the C<sub>6</sub>+ alcohols were observed (Fig. 7b), implying that NiSn@C/TiO<sub>2</sub>-1 can be used as a stable catalyst for aqueous ethanol upgradation. The contact angle of the spent NiSn@C/TiO<sub>2</sub>-1 after 5 cycle runs was 39.7° (Fig. 7c), which is close to that of the fresh one (41.1°, Fig. 3e). The amphipathicity of NiSn@C/TiO<sub>2</sub>-1 after 5 cycle runs was further validated by the optical microscope image, where the appearance of emulsification (Fig. 8) was clearly observed, which strongly confirmed the catalytic effect of emulsion to enhance the crossing-coupling of ethanol and C<sub>4</sub>+ alcohols to C<sub>6</sub>+ alcohols (Yu et al., 2013). The phase distribution of NiSn@C/TiO<sub>2</sub>-1 catalyst during stability test was also observed. As shown in Fig. 7d–h, NiSn@C/TiO<sub>2</sub>-1 catalyst exhibited excellent stability for the catalytic test in emulsion during five consecutive cycles. Additionally, the spent NiSn@C/TiO<sub>2</sub>-1 catalyst was further characterized. As shown in Fig. 9 and Fig. S4&5, the morphology, size, structure and surface properties of the spent NiSn@C/TiO<sub>2</sub>-1 also kept stable even after 5 consecutive cycles, further confirming that NiSn@C/TiO<sub>2</sub>-1 can serve as highly active, selective and stable catalyst for aqueous ethanol upgrading to C<sub>6</sub>+ alcohols.

To further confirm the role of carbon shell and TiO<sub>2</sub>, the XPS, X-ray



**Fig. 7.** Stability test for NiSn@C/TiO<sub>2</sub>-1 catalyst. (a) Ethanol conversion and liquid products selectivity, (b) carbon number distributions of liquid products, (c) the contact angle of the spent NiSn@C/TiO<sub>2</sub>-1 after 5 cycles, (d–h) distribution of catalyst in the reaction solution after each cycle (left: with catalyst, right: without catalyst). Reaction conditions:  $m_{\text{NaOH}} = 0.5$  g,  $m_{\text{catalyst}} = 0.5$  g,  $m_{\text{ethanol}} = 10$  g,  $m_{\text{water}} = 10$  g,  $T = 250$  °C, and  $t = 18$  h. The catalyst was recycled through centrifugation (Detailed method is shown in the Supporting Information).



**Fig. 8.** The optical microscope image of NiSn@C/TiO<sub>2</sub>-1 collected from the reaction system after 5 cycles.

absorption near-edge structure spectroscopy (XANES) and extended X-ray absorption fine structure spectroscopy (EXAFS) were performed to investigate the surface electronic structure and coordination environment of NiSn@C/TiO<sub>2</sub>. XPS spectra indicated no obvious difference in the surface electronic structure of NiSn@C/TiO<sub>2</sub> with different C/TiO<sub>2</sub> ratios (Fig. 10 and Table S2, S3). As shown in Fig. 11a, the peaks at  $\sim 285$ ,  $\sim 288$ , and  $\sim 292.1$  eV in the C K-edge XANES spectrum can be ascribed to the  $\pi^*$  state of C ( $sp^2$ ), C = O, and the  $\sigma^*$  state of C ( $sp^3$ ), respectively (Fu et al., 2017; Wu et al., 2021b). It is noted that the peak intensity of the  $\sigma^*$  state of C ( $sp^3$ ) of C NiSn@C/TiO<sub>2</sub>-1 is much lower

than that of graphite, suggesting that C atoms in NiSn@C/TiO<sub>2</sub>-1 are hardly coordinated with Ni (or Sn) to form C–Ni (or C–Sn) (Bare et al., 2017; Denny et al., 2014; Li et al., 2019). Moreover, Ni K-edge XANES was collected to reveal the local structure of Ni in NiSn@C/TiO<sub>2</sub>-1. It is found that the XANES spectrum of NiSn@C/TiO<sub>2</sub>-1 at Ni K-edge displays similar features to those of Ni foil, suggesting that Ni presents as metal state in NiSn@C/TiO<sub>2</sub>-1 (Fig. 11b), which is consistent with XRD and XPS results. In EXAFS spectra, the peak at  $\sim 2.5$  Å is ascribed to Ni–Ni coordination, while the absence of Ni–C coordination peak in the EXAFS spectrum of NiSn@C/TiO<sub>2</sub>-1 further confirms that Ni atoms are hardly coordinated with C atoms in NiSn@C/TiO<sub>2</sub>-1 (Fig. 11c and Table S4) (Nasu et al., 2004). In addition, the presence of Ni–Ni and the absence of Ni–C coordination in NiSn@C/TiO<sub>2</sub>-1 have been further validated by the wavelet transform contour plot (Fig. 11d) (Fernando de Souza et al., 2003; He et al., 2022; Wang et al., 2014). These results clearly confirm that the modification of carbon shell and TiO<sub>2</sub> on the NiSn@C/TiO<sub>2</sub>-1 catalyst has negligible influence on the surface chemical and electronic structure of Ni–Sn composite nanoparticles. The enhanced catalytic performance of aqueous ethanol upgrading to C<sub>6+</sub> alcohols is originated from the modulation of surface amphipathicity of catalyst.

### 3.4. Catalytic mechanism of amphipathic NiSn@C/TiO<sub>2</sub> catalyst

Higher alcohols could be produced via self-condensation of ethanol, which mainly includes three steps: 1) dehydrogenation of ethanol to acetaldehyde over metallic sites; following 2) aldol condensation of ethanol and acetaldehyde to form unsaturated aldehyde and one molecule of water over basic sites; and then 3) hydrogenation of the unsaturated aldehyde to generate higher alcohols over metallic sites. This process is hydrogen-neutral that no external hydrogen is required and the first dehydrogenation step of ethanol to acetaldehyde over metal site is recognized as the rate-limiting step (Cui et al., 2022; Davies et al.,

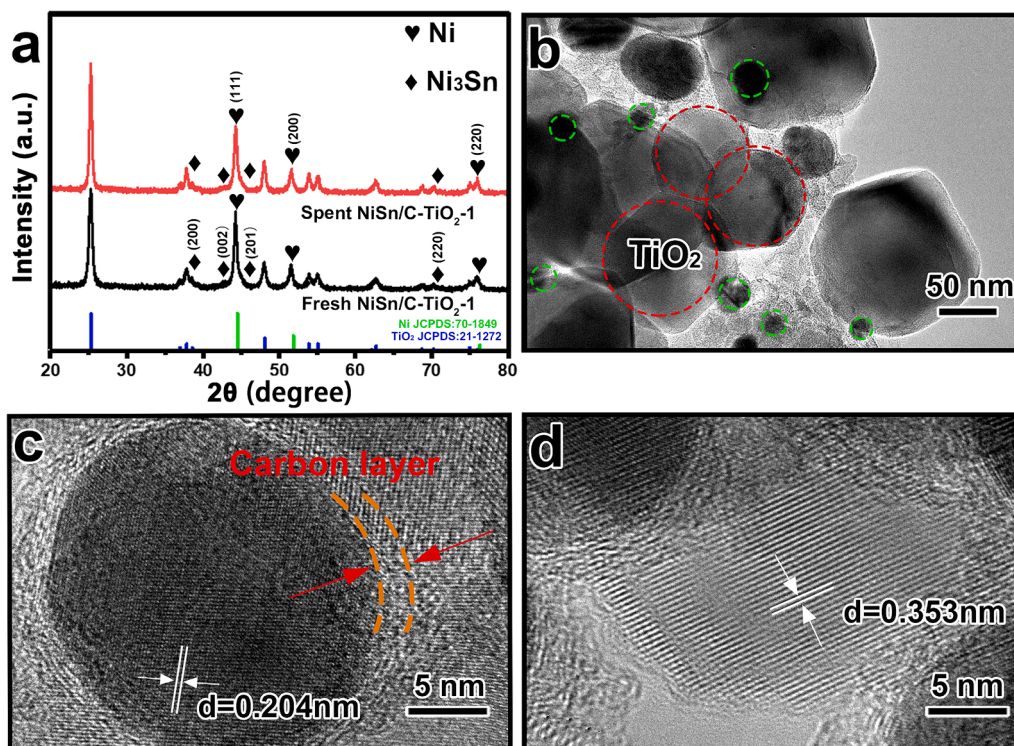


Fig. 9. (a) XRD patterns of the fresh and spent NiSn@C/TiO<sub>2</sub>-1 catalyst. (b) TEM, and (c, d) HRTEM images of the spent NiSn@C/TiO<sub>2</sub>-1 catalyst.

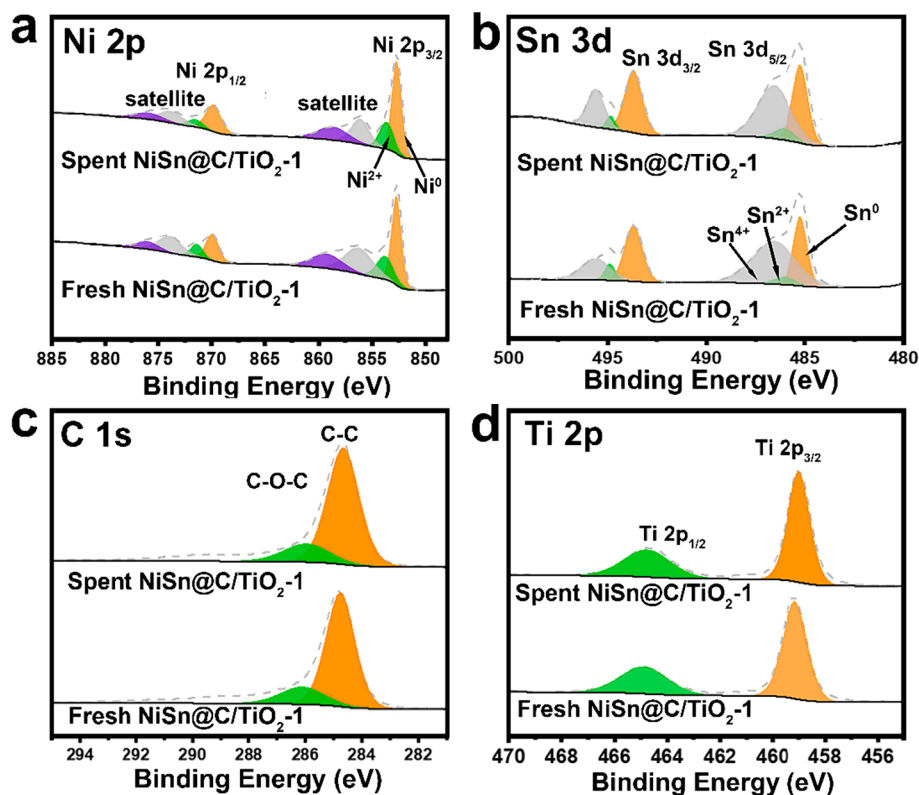
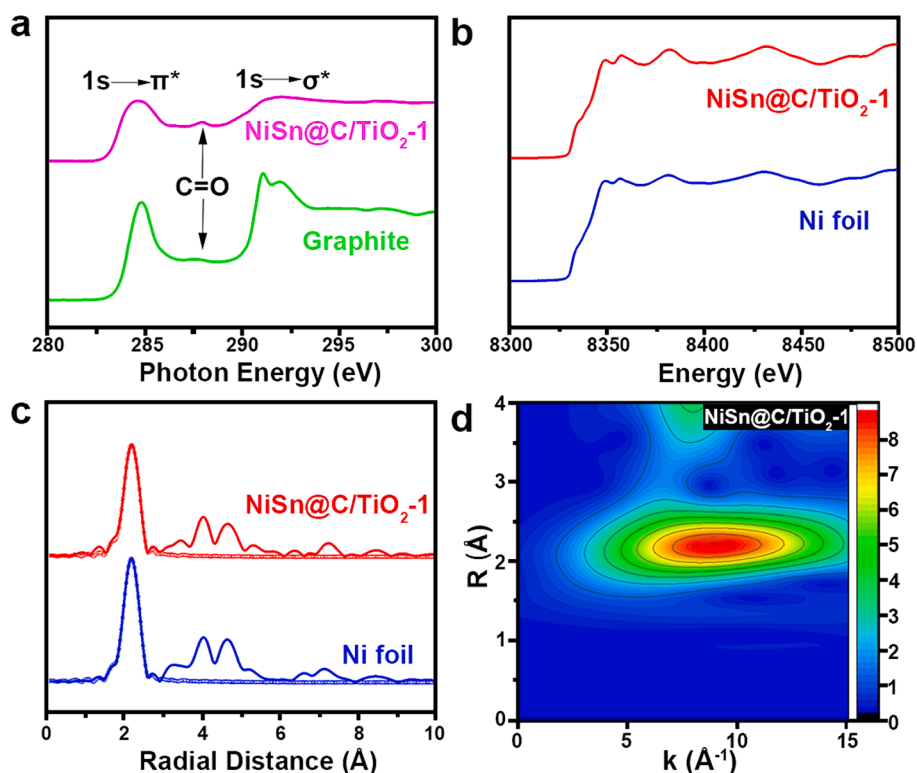


Fig. 10. XPS spectra of the fresh and spent NiSn@C/TiO<sub>2</sub>-1 catalyst: (a) Ni 2p spectra, (b) Sn 3d spectra, (c) C 1s spectra, and (d) Ti 2p spectra.

2022). Adsorption-desorption properties of NiSn/TiO<sub>2</sub>, NiSn@C/TiO<sub>2</sub>-1 and NiSn@C for ethanol, acetaldehyde and butyraldehyde were explored to give insight into their performance differences. The C<sub>2</sub>H<sub>5</sub>OH, CH<sub>3</sub>CHO and C<sub>3</sub>H<sub>7</sub>CHO-TPD are shown in Fig. 12a–c and

Fig. S6. The signals with *m/z* of 16, 46, 44 and 72 were derived from molecular ion of CH<sub>4</sub>, C<sub>2</sub>H<sub>5</sub>OH, CH<sub>3</sub>CHO and CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CHO, respectively. The oxygenated chemicals (C<sub>2</sub>H<sub>5</sub>OH, CH<sub>3</sub>CHO and C<sub>3</sub>H<sub>7</sub>CHO) are supposed to mainly adsorb on the oxygen-rich supports of



**Fig. 11.** XANES spectra at (a) C K-edge and (b) Ni K-edge. (c) EXAFS spectra of NiSn@C/TiO<sub>2</sub>-1 and Ni foil at Ni K-edge. (d) Wavelet transform contour plot of NiSn@C/TiO<sub>2</sub>-1 at Ni K-edge.

catalysts, which then could transfer onto the active metallic sites for reaction (Zhang et al., 2017). Weak adsorption of ethanol was observed on all the catalysts, with desorption peaks < 100 °C (Fig. 12a). The lowest desorption peak on the NiSn@C/TiO<sub>2</sub>-1 (87 °C for NiSn@C/TiO<sub>2</sub>-1, 96 °C for NiSn/TiO<sub>2</sub> and 100 °C for NiSn@C) indicates the weakest adsorption of ethanol on the C/TiO<sub>2</sub>-1 support and potentially the easiest transfer of ethanol onto the Ni sites for the rate-limiting dehydrogenation reaction. Obvious signal of CH<sub>4</sub> appeared at 329 °C, 306 °C and 314 °C over NiSn@C, NiSn@C/TiO<sub>2</sub>-1 and NiSn/TiO<sub>2</sub> catalysts, respectively (Fig. S6). The lowest desorption peak of CH<sub>4</sub> on the NiSn@C/TiO<sub>2</sub>-1 demonstrates the minimum methanation activity. The adsorption properties of the intermediates, i.e. acetaldehyde and butyraldehyde on the catalysts in reverse could show their transfer potential from the catalysts surface after production into to the reaction solution. From CH<sub>3</sub>CHO and C<sub>3</sub>H<sub>7</sub>CHO-TPD in Fig. 12b&c, the lowest desorption peak of both CH<sub>3</sub>CHO and C<sub>3</sub>H<sub>7</sub>CHO on the NiSn@C/TiO<sub>2</sub>-1 catalyst implies their weakest adsorption and easiest desorption into the reaction solvent. The prompt desorption of the aldehyde intermediates from the NiSn@C/TiO<sub>2</sub>-1 catalyst effectively inhibited their overreaction on the catalyst and promoted their subsequent aldol-type C-C coupling by catalysis of alkali in the aqueous phase for higher alcohols production.

Due to the hydrophobic properties of C<sub>4+</sub> higher alcohols, a separate oil phase gradually formed in the aqueous reaction system with formation of more and more C<sub>4+</sub>, especially C<sub>6+</sub> higher alcohols. As atradiational heterogeneous catalyst is normally either hydrophilic or oleophilic, separated into aqueous phase or oil phase, the cross-coupling of ethanol in aqueous phase and butanol or C<sub>6+</sub> alcohols in oil phase is usually blocked. To break through this limitation, a hybrid NiSn@C/TiO<sub>2</sub> catalyst which could work stably in the oil/water interface was developed by combining hydrophilic TiO<sub>2</sub> and oleophilic carbon shells. NiSn NPs were used as metal active center and TiO<sub>2</sub> acted as the hydrophilic support, while carbon shells as lipophilic support protected NiSn nanoparticles from agglomeration. The catalysis process by the

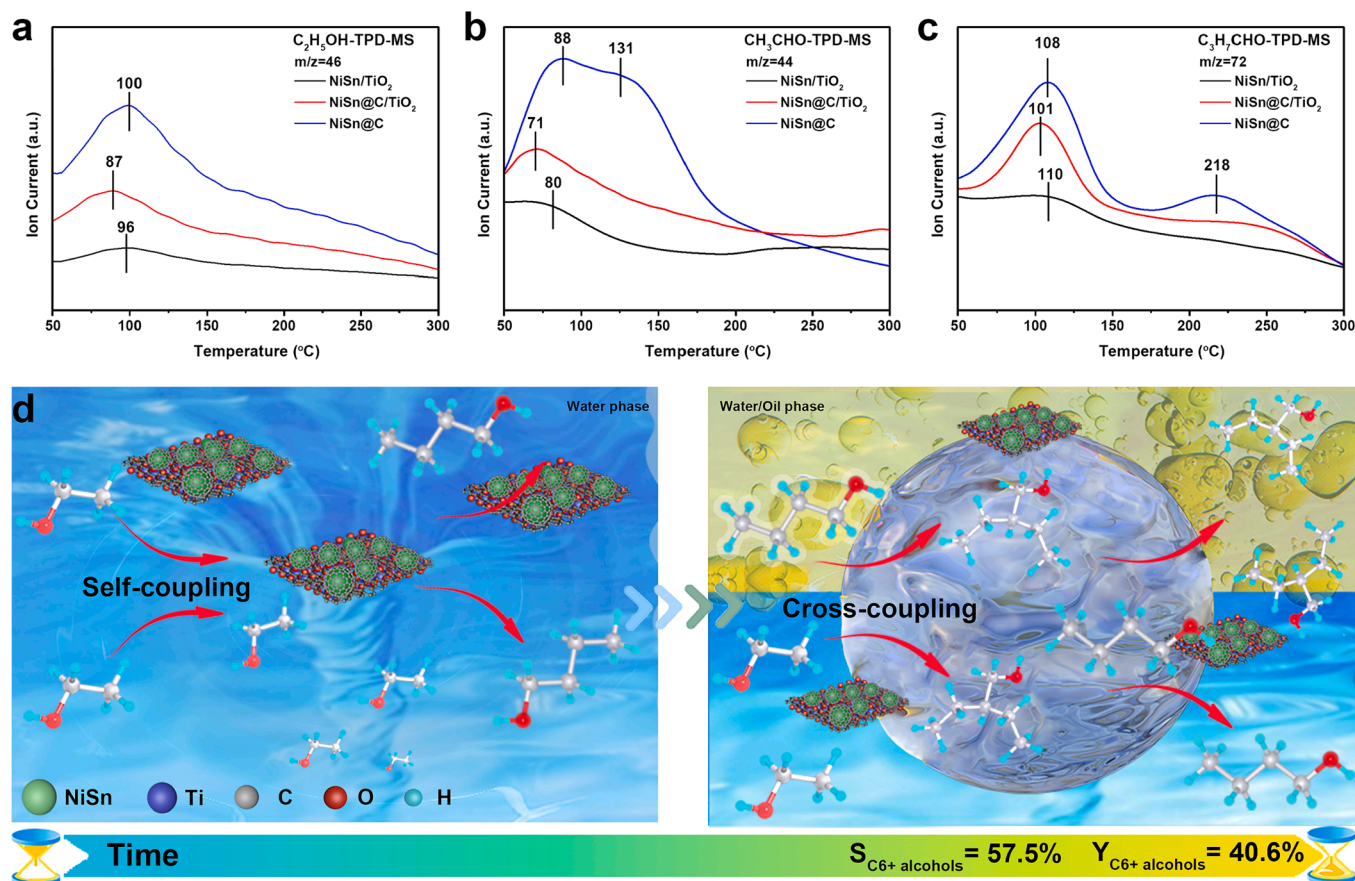
NiSn@C/TiO<sub>2</sub> catalyst could be described as Fig. 12d. There was only water phase containing water and ethanol at the initial stage of the reaction. With production of C<sub>4+</sub> alcohols, oil-in-water emulsion stabilized by the amphipathic NiSn@C/TiO<sub>2</sub> catalyst gradually formed. Accordingly, the cross-coupling reaction was facilitated by the easily gathering and reaction of the water-soluble substrates (C<sub>2</sub>-C<sub>4</sub>, etc.) and oil-soluble intermediates (C<sub>4+</sub>, etc.) over the amphipathic NiSn@C/TiO<sub>2</sub> catalyst in the interface of the emulsion. As a result, the chain growth was promoted and the yield of C<sub>6+</sub> higher alcohols was significantly increased on the NiSn@C/TiO<sub>2</sub> catalyst.

#### 4. Conclusion

In summary, we have demonstrated that the regulation of the hydrophilic and lipophilic properties of catalysts can significantly boost the performance towards ethanol upgrading to C<sub>6+</sub> alcohols. Detailed characterization and experimental observations confirm that the modification of NiSn NPs with carbon shell and TiO<sub>2</sub> precisely regulates the amphipathicity, which acts as interface catalyst to promote the transformation from self-coupling of aqueous ethanol to crossing-coupling of ethanol and higher alcohols at the oil/water interface. The optimal NiSn@C/TiO<sub>2</sub>-1 catalyst exhibits excellent performance with ethanol conversion of 70.6 % and C<sub>6+</sub> alcohols selectivity of 57.5 %, accounting for C<sub>6+</sub> alcohols yield of 40.6 %, which is 2–3 folds higher than that of the reported works (Table S5). This work may not only provide a novel approach for the C<sub>6+</sub> alcohols production by direct upgrading of aqueous ethanol from biomass fermentation, but also promote the fundamental researches on the design of phase transfer catalysts for the catalytic effect of emulsion.

#### CRediT authorship contribution statement

**Xiaoping Wu:** Conceptualization, Data curation, Funding acquisition, Methodology, Writing – original draft, Writing – review & editing.



**Fig. 12.** (a) C<sub>2</sub>H<sub>5</sub>OH-TPD-MS profiles, (b) CH<sub>3</sub>CHO-TPD-MS profiles, and (c) C<sub>3</sub>H<sub>7</sub>CHO-TPD-MS profiles of NiSn/TiO<sub>2</sub>, NiSn@C/TiO<sub>2</sub>-1 and NiSn@C. (d) Schematic illustration of the ethanol upgrading reaction pathway over NiSn@C/TiO<sub>2</sub>-1 catalyst.

**Guohao Zou:** Data curation, Formal analysis, Methodology, Writing – original draft. **Jiaying Hou:** Data curation, Methodology. **Xinrui Liu:** Data curation, Formal analysis. **Mingge Li:** Data curation, Formal analysis. **Xueying Cai:** Data curation, Methodology. **Qian Zhang:** Methodology. **Yunhong Pi:** Writing – review & editing. **Qingwei Meng:** Formal analysis, Methodology, Writing – review & editing. **Tiejun Wang:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Writing – review & editing.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Data availability

No data was used for the research described in the article.

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#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ces.2024.120013>.

#### References

- Aitchison, H., Wingad, R.L., Wass, D.F., 2016. Homogeneous ethanol to butanol catalysis—Guerbet renewed. *ACS Catal.* 6, 7125–7132. <https://doi.org/10.1021/acscatal.6b01883>.
- Baggetto, L., Ganesh, P., Meisner, R.P., Unocic, R.R., Jumas, J.-C., Bridges, C.A., Veith, G. M., 2013. Characterization of sodium ion electrochemical reaction with tin anodes: Experiment and theory. *J. Power Sources* 234, 48–59. <https://doi.org/10.1016/j.jpowsour.2013.01.083>.
- Bare, S.R., Vila, F.D., Charochak, M.E., Prabhakar, S., Bradley, W.J., Jaye, C., Fischer, D. A., Hayashi, S.T., Bradley, S.A., Rehr, J.J., 2017. Characterization of coke on a Pt-Re/ $\gamma$ -Al<sub>2</sub>O<sub>3</sub> re-forming catalyst: experimental and theoretical study. *ACS Catal.* 7, 1452–1461. <https://doi.org/10.1021/acscatal.6b02785>.
- Crossley, S., Faria, J., Shen, M., Resasco, D.E., 2010. Solid nanoparticles that catalyze biofuel upgrade reactions at the water/oil interface. *Science* 327, 68–72. <https://doi.org/10.1126/science.1180769>.
- Cui, Y., Li, S., An, H., Zhao, X., Wang, Y., 2022. Improvement of ethanol guerbet condensation by acetal hydrolysis. *Ind. Eng. Chem. Res.* 61, 12392–12404. <https://doi.org/10.1021/acs.iecr.2c01382>.
- Davies, A.M., Li, Z.-Y., Stephenson, C.R.J., Szymczak, N.K., 2022. Valorization of ethanol: ruthenium-catalyzed guerbet and sequential functionalization processes. *ACS Catal.* 12, 6729–6736. <https://doi.org/10.1021/acscatal.2c01570>.
- De Pours, M.V., Gopal, K., Sathiyagnanam, A.P., Rajesh Kumar, B., Rana, D., Saravanan, S., Damodharan, D., 2020. Comparative account of the effects of two high carbon alcohols (C5 & C6) on combustion, performance and emission characteristics of a DI diesel engine. *Energy Sources Part A* 42, 1772–1784. <https://doi.org/10.1080/15567036.2019.1604888>.
- Denny, Y.R., Takahashi, K., Niki, K., Yang, D.S., Fujikawa, T., Kang, H.J., 2014. Ni K-edge XAFS analysis of NiO thin film with multiple scattering theory. *Surf. Interface Anal.* 46, 997–999. <https://doi.org/10.1002/sia.5455>.
- Eagan, N.M., Moore, B.M., McClelland, D.J., Wittrig, A.M., Canales, E., Lanci, M.P., Huber, G.W., 2019. Correction: catalytic synthesis of distillate-range ethers and

- olefins from ethanol through guerbet coupling and etherification. *Green Chem.* 21, 5128–5129. <https://doi.org/10.1039/C9GC90076D>.
- Earley, J.H., Bourne, R.A., Watson, M.J., Poliakov, M., 2015. Continuous catalytic upgrading of ethanol to n-butanol and >C4 products over Cu/CeO<sub>2</sub> catalysts in supercritical CO<sub>2</sub>. *Green Chem.* 17, 3018–3025. <https://doi.org/10.1039/C4GC00219A>.
- Fernando de Souza, R., Simon, L.C., Alves, M.d.C.M., 2003. XAS study of the nickel ( $\alpha$ -diimine) catalyst for olefin polymerization. *J. Catal.* 214, 165–168. [https://doi.org/10.1016/S0021-9517\(02\)00137-9](https://doi.org/10.1016/S0021-9517(02)00137-9).
- Fu, S., Shao, Z., Wang, Y., Liu, Q., 2017. Manganese-catalyzed upgrading of ethanol into 1-butanol. *J. Am. Chem. Soc.* 139, 11941–11948. <https://doi.org/10.1021/jacs.7b05939>.
- Galadima, A., Muraza, O., 2015. Catalytic upgrading of bioethanol to fuel grade biobutanol: a review. *Ind. Eng. Chem. Res.* 54, 7181–7194. <https://doi.org/10.1021/acs.iecr.5b01443>.
- Goh, B.H.H., Chong, C.T., Ong, H.C., Seljak, T., Katrašnik, T., Józsa, V., Ng, J.-H., Tian, B., Karmarkar, S., Ashokkumar, V., 2022. Recent advancements in catalytic conversion pathways for synthetic jet fuel produced from bioresources. *Energy Convers. Manage.* 251, 114974. <https://doi.org/10.1016/j.enconman.2021.114974>.
- Han, G.-F., Chen, Z.-W., Jeon, J.-P., Kim, S.-J., Noh, H.-J., Shi, X.-M., Li, F., Jiang, Q., Baek, J.-B., 2019. Low-temperature conversion of alcohols into bulky nanoporous graphene and pure hydrogen with robust selectivity on CaO. *Adv. Mater.* 31, 1807267. <https://doi.org/10.1002/adma.201807267>.
- Han, X., Li, S., An, H., Zhao, X., Wang, Y., 2021. Improvement of n-butanol guerbet condensation: a reaction integration of n-butanol guerbet condensation and 1,1-dibutoxybutane hydrolysis. *React. Chem. Eng.* 6, 1845–1853. <https://doi.org/10.1039/D1RE00206F>.
- He, J., Li, X., Kou, J., Tao, T., Shen, X., Jiang, D., Lin, L., Li, X., 2023. Catalytic upgrading of ethanol to higher alcohols over nickel-modified Cu-LazO<sub>3</sub>/Al<sub>2</sub>O<sub>3</sub> catalysts. *Cat. Sci. Technol.* 13, 170–177. <https://doi.org/10.1039/D2CY01554D>.
- He, W., Liu, H., Cheng, J., Li, Y., Liu, C., Chen, C., Zhao, J., Xin, H.L., 2022. Modulating the electronic structure of nickel sulfide electrocatalysts by chlorine doping toward highly efficient alkaline hydrogen evolution. *ACS Appl. Mater. Interfaces* 14, 6869–6875. <https://doi.org/10.1021/acsami.1c23251>.
- Karpagam, R., Rani, K., Ashokkumar, B., Ganesh Moorthy, I., Dhakshinamoorthy, A., Varalakshmi, P., 2020. Green energy from Coelastrella sp. M-60: bio-nanoparticles mediated whole biomass transesterification for biodiesel production. *Fuel* 279, 118490. <https://doi.org/10.1016/j.fuel.2020.118490>.
- Kozłowski, J.T., Davis, R.J., 2013. Heterogeneous catalysts for the guerbet coupling of alcohols. *ACS Catal.* 3, 1588–1600. <https://doi.org/10.1021/cs400292f>.
- Kulkarni, N.V., Brennessel, W.W., Jones, W.D., 2018. Catalytic upgrading of ethanol to n-butanol via manganese-mediated guerbet reaction. *ACS Catal.* 8, 997–1002. <https://doi.org/10.1021/acscatal.7b03653>.
- Li, D., Xu, C., Yeager, C.M., Lin, P., Xing, W., Schwehr, K.A., Chen, N., Arthur, Z., Kaplan, D.I., Santschi, P.H., 2019. Molecular interaction of aqueous iodine species with humic acid studied by I and C K-edge X-ray absorption spectroscopy. *Environ. Sci. Tech.* 53, 12416–12424. <https://doi.org/10.1021/acs.est.9b03682>.
- Liao, J., Zhong, Q., Gu, J., Qiu, S., Meng, Q., Zhang, Q., Wang, T., 2022. New approach for bio-jet fuels production by hydrodeoxygenation of higher alcohols derived from C-C coupling of bio-ethanol. *Appl. Energy* 324, 119843. <https://doi.org/10.1016/j.apenergy.2022.119843>.
- Liu, Y., Shao, Z., Wang, Y., Xu, L., Yu, Z., Liu, Q., 2019. Manganese-catalyzed selective upgrading of ethanol with methanol into isobutanol. *ChemSusChem* 12, 3069–3072. <https://doi.org/10.1002/cssc.201802689>.
- Luo, Y., Zhang, R., He, Y., Lou, D., Zhu, R., Zhu, C., Fan, M., Li, Q., 2023a. Preparation of bio-jet fuels by a controllable integration process: coupling of biomass fermentation and olefin polymerization. *Bioresour. Technol.* 382, 129175. <https://doi.org/10.1016/j.biortech.2023.129175>.
- Luo, Y., He, Y., Zhang, R., Lou, D., Zhu, R., Zhu, C., Li, Q., 2023b. A novel integrated process for selective preparation of jet-range fuels from low carbon alcohols and ABE. *Fuel Process. Technol.* 241, 107603. <https://doi.org/10.1016/j.fuproc.2022.107603>.
- Mika, L.T., Cséfalvay, E., Németh, Á., 2018. Catalytic conversion of carbohydrates to initial platform chemicals: chemistry and sustainability. *Chem. Rev.* 118, 505–613. <https://doi.org/10.1021/acs.chemrev.7b00395>.
- Morganti, K., Almansour, M., Khan, A., Kalghatgi, G., Przesmitzki, S., 2018. Leveraging the benefits of ethanol in advanced engine-fuel systems. *Energy Convers. Manage.* 157, 480–497. <https://doi.org/10.1016/j.enconman.2017.11.086>.
- Nasu, T., Sakurai, M., Kamiyama, T., Usuki, T., Uemura, O., Tokumitsu, K., Yamasaki, T., 2004. Structural comparison of M-W (M = Fe, Ni) alloys produced by electrodeposition and mechanical alloying. *Mater. Sci. Eng. A* 375–377, 163–170. <https://doi.org/10.1016/j.msea.2003.10.144>.
- Pang, J., Zheng, M., He, L., Li, L., Pan, X., Wang, A., Wang, X., Zhang, T., 2016. Upgrading ethanol to n-butanol over highly dispersed Ni-MgAlO catalysts. *J. Catal.* 344, 184–193. <https://doi.org/10.1016/j.jcat.2016.08.024>.
- Pang, J., Yin, M., Wu, P., Li, X., Li, H., Zheng, M., Zhang, T., 2021. Advances in catalytic dehydrogenation of ethanol to acetaldehyde. *Green Chem.* 23, 7902–7916. <https://doi.org/10.1039/D1GC02799A>.
- Pérez, L.E.A., Casas, Y., Peralta, L.M., Prieto, J.O., Granda, D., Tretó, M.A., 2011. Assessment of the energy quality of the synthesis gas produced from biomass derived fuels conversion: Part I: liquid fuels, ethanol. *Afinidad* 68, 285–289.
- Rorrer, J., Pindi, S., Toste, F.D., Bell, A.T., 2018. Effect of alcohol structure on the kinetics of etherification and dehydration over tungstated zirconia. *ChemSusChem* 11, 3104–3111. <https://doi.org/10.1002/cssc.201801067>.
- Shields, S., Boopathy, R., 2011. Ethanol production from lignocellulosic biomass of energy cane. *Int Biodegr Biodegr* 65, 142–146. <https://doi.org/10.1016/j.ibiod.2010.10.006>.
- Usha, K.S., Sivakumar, R., Sanjeeviraja, C., 2022. Influence of substrate temperature on the optical and vibrational properties of RF sputter coated NiO:WO<sub>3</sub> thin films for optoelectronic applications. *Opt. Mater.* 132, 112847. <https://doi.org/10.1016/j.optmat.2022.112847>.
- Wang, J., Yang, W., Wu, C., Gong, Y., Zhang, J., Shen, C., 2020a. Upgrading n-butanol to branched alcohols over Ni/Ca<sub>2</sub>Mg<sub>2</sub>O. *ACS Sustain. Chem. Eng.* 8, 16960–16967. <https://doi.org/10.1021/acssuschemeng.0c07070>.
- Wang, S., Zhang, D., Ma, Y., Zhang, H., Gao, J., Nie, Y., Sun, X., 2014. Aqueous solution synthesis of Pt-M (M = Fe, Co, Ni) bimetallic nanoparticles and their catalysis for the hydrolytic dehydrogenation of ammonia borane. *ACS Appl. Mater. Interfaces* 6, 12429–12435. <https://doi.org/10.1021/am502335j>.
- Wang, K., Zhang, L., Tang, W., Sun, H., Xue, D., Lei, M., Xiao, J., Wang, C., 2020b. Asymmetric guerbet reaction to access chiral alcohols. *Angew. Chem. Int. Ed.* 59, 11408–11415. <https://doi.org/10.1002/anie.202003104>.
- Wu, X., Cai, X., Zhang, Q., Bi, P., Meng, Q., Pi, Y., Wang, T., 2021a. Upgrading of aqueous bioethanol to higher alcohols over NiSn/MgAlO catalyst. *ACS Sustain. Chem. Eng.* 9, 11269–11279. <https://doi.org/10.1021/acssuschemeng.1c04305>.
- Wu, Z., Wang, P., Wang, J., Tan, T., 2021b. Guerbet reactions for biofuel production from ABE fermentation using bifunctional ni-MgO-Al<sub>2</sub>O<sub>3</sub> catalysts. *Catalysts* 11, 414. <https://doi.org/10.3390/catal11040414>.
- Yu, C., Fan, L., Yang, J., Shan, Y., Qiu, J., 2013. Phase-reversal emulsion catalysis with CNT-TiO<sub>2</sub> nanohybrids for the selective oxidation of benzyl alcohol. *Chem. -Eur. J.* 19, 16192–16195. <https://doi.org/10.1002/chem.201300949>.
- Yuan, B., Zhang, J., An, Z., Zhu, Y., Shu, X., Song, H., Xiang, X., Wang, W., Jing, Y., Zheng, L., He, J., 2022. Atomic Ru catalysis for ethanol coupling to C4+ alcohols. *Appl Catal B* 309, 121271. <https://doi.org/10.1016/j.apcatb.2022.121271>.
- Zhang, Q., Dong, J., Liu, Y., Wang, Y., Cao, Y., 2016. Towards a green bulk-scale biobutanol from bioethanol upgrading. *J. Energy Chem.* 25, 907–910. <https://doi.org/10.1016/j.jechem.2016.08.010>.
- Zhang, R., He, Y., Luo, Y., Lou, D., Zhu, R., Zhu, C., Li, Q., 2023. Selective preparation of jet fuels from low carbon alcohols and ABE at atmospheric pressure. *Energy* 281, 128246. <https://doi.org/10.1016/j.energy.2023.128246>.
- Zhang, P., Wang, Q.-N., Yang, X., Wang, D., Li, W.-C., Zheng, Y., Chen, M., Lu, A.-H., 2017. A highly porous carbon support rich in graphitic-N stabilizes copper nanocatalysts for efficient ethanol dehydrogenation. *ChemCatChem* 9, 505–510. <https://doi.org/10.1002/cctc.201601373>.
- Zhao, J., Xu, Y., Wang, W., Griffin, J., Roozeboom, K., Wang, D., 2020. Bioconversion of industrial hemp biomass for bioethanol production: a review. *Fuel* 281, 118725. <https://doi.org/10.1016/j.fuel.2020.118725>.
- Zhu, Z., Ding, D., Zhang, Y., Zhang, Y., 2020. Preparation of Ni, CoO-supported hollows nanotube catalyst and its application in the hydrogenation of furfural to furfuryl alcohol. *Appl. Clay Sci.* 196, 105761. <https://doi.org/10.1016/j.clay.2020.105761>.